

Safeguard-The high-Security Mission

Text-Based Adventure Game using C++



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Computer Programming Project

Project Documentation

1) BACKGROUND

1.1) Introduction

This project is a text-based adventure game called "Safeguard: The High-Security Mission." The player takes on the role of the head of security at a top-secret research facility and must stop an intruder from accessing sensitive research files.

1.2) Project Idea

The idea is to create an interactive game where the player makes choices that affect the outcome of the story. The game is designed to be engaging and challenging, with multiple paths and endings based on the player's decisions.

1.3) Solution

The solution is implemented in C++ and uses simple input/output operations to guide the player through the adventure. The game logic uses conditional statements to handle different choices and outcomes.

2) PROJECT SCOPE

The project scope includes:

- Developing a text-based adventure game.
- Implementing multiple decision points and outcomes.
- Ensuring the game is engaging and challenging.

Code:

```
#include <iostream>
```

```
#include <string>
```

```
using namespace std;
```

```
// Function to handle the main part of the adventure
```

```
void startAdventure() {
```

```
    string choice;
```

```
    cout << "\n Welcome to Safeguard: The High-Security  
Mission\n";
```

```
    cout << "\nAn alarm sounds. You are the head of security  
in a University at a top-secret research facility.\n";
```

```
    cout << "Your mission is to stop an intruder from accessing  
the university's sensitive research files.\n";
```

```
    cout << "\nYou are currently in the security office. There  
are 3 areas you can go to:\n";
```

```
    cout << "1. Control Room (for monitoring security cameras  
and alarms)\n";
```

```
    cout << "2. Elevator (to head to the top floor where the  
files are stored)\n";
```

```
cout << "3. Storage Room (to gather supplies)\n";  
  
cout << "\nWhat will you do? (Enter 1, 2, or 3): ";  
  
cin >> choice;  
  
if (choice == "1") {  
  
    cout << "\nYou enter the Control Room. The screens  
show the intruder has reached the main lobby.\n";  
  
    cout << "You can:\n";  
  
    cout << "1. Lock all doors to slow the intruder.\n";  
  
    cout << "2. Use the intercom system to alert all  
personnel.\n";  
  
    cout << "\nWhat will you do? (Enter 1 or 2): ";  
  
    cin >> choice;  
  
    if (choice == "1") {  
  
        cout << "\nThe doors are locked. The intruder is  
delayed, but they are trying to disable the locks.\n";  
  
        cout << "Do you want to:\n";  
  
        cout << "1. Stay in the Control Room and monitor  
their progress.\n";  
  
        cout << "2. Head to the Elevator to prepare for a direct  
confrontation.\n";  
  
        cout<<"\n What will you do?(Enter 1 or 2) :";
```

```
cin >> choice;

if (choice == "1") {

    cout << "\nYou monitor the screens. The intruder
disables the locks faster than expected and heads to the
elevator.\n";

    cout << "You rush to the top floor but are too late.
Game Over.\n";

} else {

    cout << "\nYou head to the Elevator. You intercept
the intruder and, after a tense standoff, manage to stop
them.\n";

    cout << "Mission accomplished!\n";

}

} else {

    cout << "\nYou alert all personnel. A team rushes to
intercept the intruder, but they are overpowered.\n";

    cout << "The intruder reaches the elevator and
disables the alarms. Game Over.\n";

}

}

else if (choice == "2") {
```

```
cout << "\nYou head to the Elevator. The intruder is  
already in the building.\n";
```

```
cout << "Do you:\n";
```

```
cout << "1. Take the elevator to the top floor to secure  
the files?\n";
```

```
cout << "2. Wait to ambush the intruder near the  
elevator?\n";
```

```
cin >> choice;
```

```
if (choice == "1") {
```

```
    cout << "\nYou secure the top floor. The intruder,  
however, has bypassed other security measures and is heading  
toward you.\n";
```

```
    cout << "Do you:\n";
```

```
    cout << "1. Engage the intruder in a fight?\n";
```

```
    cout << "2. Use a diversion to escape with the  
files?\n";
```

```
    cin >> choice;
```

```
    if (choice == "1") {
```

```
        cout << "\nYou engage the intruder in a fight and  
manage to subdue them after a tough battle.\n";
```

```
        cout << "You have saved the research files. Mission  
accomplished!\n";
```

```
    } else {  
  
        cout << "\nYou create a diversion, but the intruder  
sees through it and catches you.\n";  
  
        cout << "Game Over.\n";  
  
    }  
  
} else {  
  
    cout << "\nYou wait near the elevator and ambush the  
intruder as they approach.\n";  
  
    cout << "After a brief struggle, you neutralize the  
threat. Mission accomplished!\n";  
  
    }  
  
}  
  
else if (choice == "3") {  
  
    cout << "\nYou enter the Storage Room to gather  
supplies: weapons, gadgets, and a first-aid kit.\n";  
  
    cout << "You hear footsteps approaching! Do you:\n";  
  
    cout << "1. Hide in the shadows?\n";  
  
    cout << "2. Confront the intruder?\n";  
  
    cin >> choice;  
  
    if (choice == "1") {
```

```
cout << "\nYou hide successfully and watch as the  
intruder passes by.\n";
```

```
cout << "Do you:\n";
```

```
cout << "1. Ambush the intruder now?\n";
```

```
cout << "2. Follow them to learn their plan?\n";
```

```
cin >> choice;
```

```
if (choice == "1") {
```

```
    cout << "\nYou ambush the intruder and manage to  
neutralize them. You have saved the day!\n";
```

```
    } else {
```

```
        cout << "\nYou follow the intruder to the elevator.  
They discover you and overpower you.\n";
```

```
        cout << "Game Over.\n";
```

```
    }
```

```
    } else {
```

```
        cout << "\nYou confront the intruder. A battle ensues,  
and you are injured but manage to stop them.\n";
```

```
        cout << "Mission accomplished, but you need medical  
attention.\n";
```

```
    }
```

```
}
```



```
else {  
    cout << "\nInvalid choice. Please restart the game and  
enter a valid option.\n";  
}  
  
}  
  
int main() {  
    string playAgain;  
    do{  
        startAdventure(); // Start the adventure game  
        cout<<"\n would u like to play again?(y/n) ";  
        cin>> playAgain;  
        cout<<"\n -----  
-----" << endl;  
    }  
    while(playAgain == "y" || playAgain == "Y");  
    cout<<"Thank you for playing!"<<endl;  
    cout<<"Good Bye!"<<endl;  
    return 0;  
}
```

3) PROPOSAL

3.1) Vision and Goals

- The vision is to create an engaging and interactive text-based adventure game.
- The goals are to provide an enjoyable gaming experience, implement multiple decision points, and ensure the game is challenging and replayable.

3.2) Software Requirements

- C++ compiler
- Text editor or IDE (e.g., Dev C++)

4) REQUIREMENTS

4.1) Functional Requirements

- The game should prompt the player for input at various decision points.
- The game should provide different outcomes based on the player's choices.
- The game should have a clear start and end.
- Validate player to ensure that only valid options are accepted
- Provide error handling for invalid inputs.(e.g. “Please enter a number between 1 and 3”)

4.2) Mapping FR Through C++

- **Input handling:** Use cin to capture player choices.
- **Decision points:** Use if-else statements to handle different choices and outcomes.
- **Game flow:** Use functions to structure the game logic and handle different parts of the adventure.

4.3) Non-Functional Requirements

- The game should be easy to understand and play.
- The game should provide clear instructions and feedback to the player.
- The game should be free of bugs and run smoothly.

5) SCHEDULE

5.1) Road Map

1. Define the game concept and outline the story.
2. Implement the basic game structure and input handling.
3. Add decision points and multiple outcomes.
4. Test the game and fix any bugs.
5. Finalize the game and prepare documentation.

6) CONCLUSION

This project aims to create an engaging text-based adventure game using C++. The game provides an interactive experience with multiple decision points and outcomes. The project demonstrates the use of basic C++ programming concepts and provides a foundation for further development and enhancements.

7) Validation and Verification

- **Validation:** Ensure the game meets the functional requirements by testing different decision paths and outcomes.
- **Verification:** Review the code to ensure it follows best practices and is free of errors.

Output

```
Welcome to Safeguard: The High-Security Mission

An alarm sounds. You are the head of security in a University at a top-secret research facility.
Your mission is to stop an intruder from accessing the university's sensitive research files.

You are currently in the security office. There are 3 areas you can go to:
1. Control Room (for monitoring security cameras and alarms)
2. Elevator (to head to the top floor where the files are stored)
3. Storage Room (to gather supplies)

What will you do? (Enter 1, 2, or 3): 1

You enter the Control Room. The screens show the intruder has reached the main lobby.
You can:
1. Lock all doors to slow the intruder.
2. Use the intercom system to alert all personnel.

What will you do? (Enter 1 or 2): 1

The doors are locked. The intruder is delayed, but they are trying to disable the locks.
Do you want to:
1. Stay in the Control Room and monitor their progress.
2. Head to the Elevator to prepare for a direct confrontation.

What will you do?(Enter 1 or 2) :1

You monitor the screens. The intruder disables the locks faster than expected and heads to the elevator.
You rush to the top floor but are too late. Game Over.

would u like to play again?(y/n) y

-----

Welcome to Safeguard: The High-Security Mission

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You are currently in the security office. There are 3 areas you can go to:
1. Control Room (for monitoring security cameras and alarms)
2. Elevator (to head to the top floor where the files are stored)
3. Storage Room (to gather supplies)

What will you do? (Enter 1, 2, or 3):
```

Output

[Clear](#)

Welcome to Safeguard: The High-Security Mission

An alarm sounds. You are the head of security in a University at a top-secret research facility. Your mission is to stop an intruder from accessing the university's sensitive research files.

You are currently in the security office. There are 3 areas you can go to:

1. Control Room (for monitoring security cameras and alarms)
2. Elevator (to head to the top floor where the files are stored)
3. Storage Room (to gather supplies)

What will you do? (Enter 1, 2, or 3): 3

You enter the Storage Room to gather supplies: weapons, gadgets, and a first-aid kit. You hear footsteps approaching! Do you:

1. Hide in the shadows?
2. Confront the intruder?

2

You confront the intruder. A battle ensues, and you are injured but manage to stop them. Mission accomplished, but you need medical attention.

would u like to play again?(y/n) n

Thank you for playing!
Good Bye!

=== Code Execution Successful ===

Output

[Clear](#)

Welcome to Safeguard: The High-Security Mission

An alarm sounds. You are the head of security in a University at a top-secret research facility. Your mission is to stop an intruder from accessing the university's sensitive research files.

You are currently in the security office. There are 3 areas you can go to:

1. Control Room (for monitoring security cameras and alarms)
2. Elevator (to head to the top floor where the files are stored)
3. Storage Room (to gather supplies)

What will you do? (Enter 1, 2, or 3): 4

Invalid choice. Please restart the game and enter a valid option.

would u like toplay again?(y/n) |